Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
    a. Show differences and similarities
        i.Map how each affects the Data structure and Security level
        ii.Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
    a. Show how each concept is mathematically connected
    b. Map their role in encryption/decryption
    c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Name:- G. Ram srinivas

Reg:- 192384068

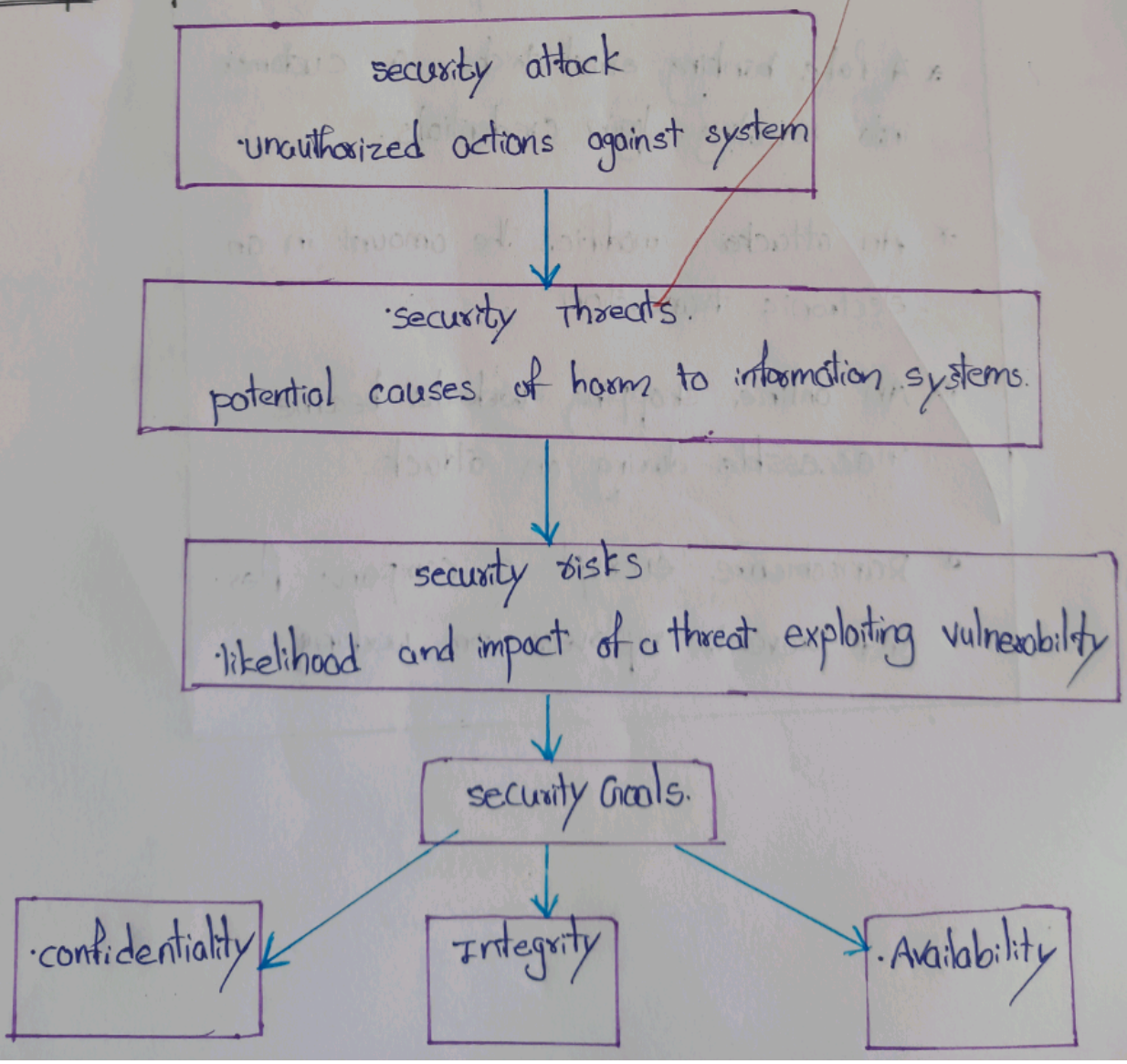
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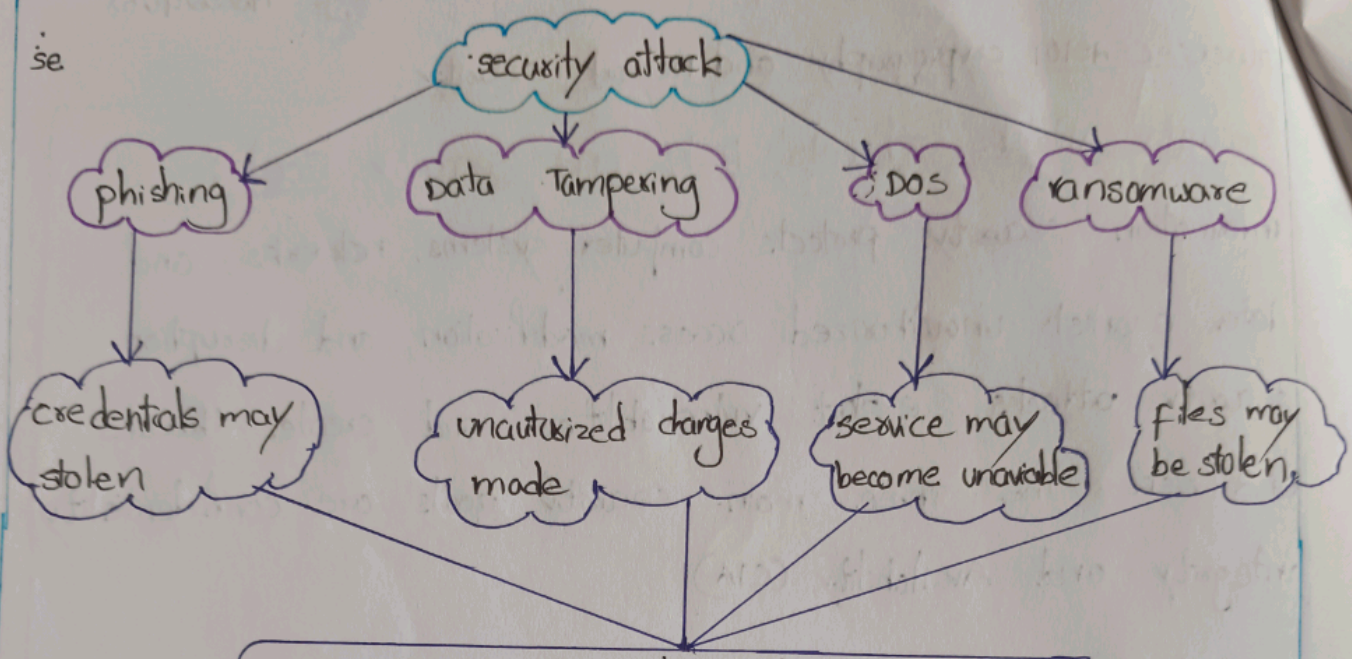
course:- CSA5101 cryptography and Network security

1. **Security Attacks, Threats, Risks and security Goals:-**

Information security protects computer systems, networks, and data aganist unauthorized access, modification, and disruption. security attacks exploit vulnerabilities and create threats and risks. the three main security goals are confidentiality integrity and Availability (CIA)

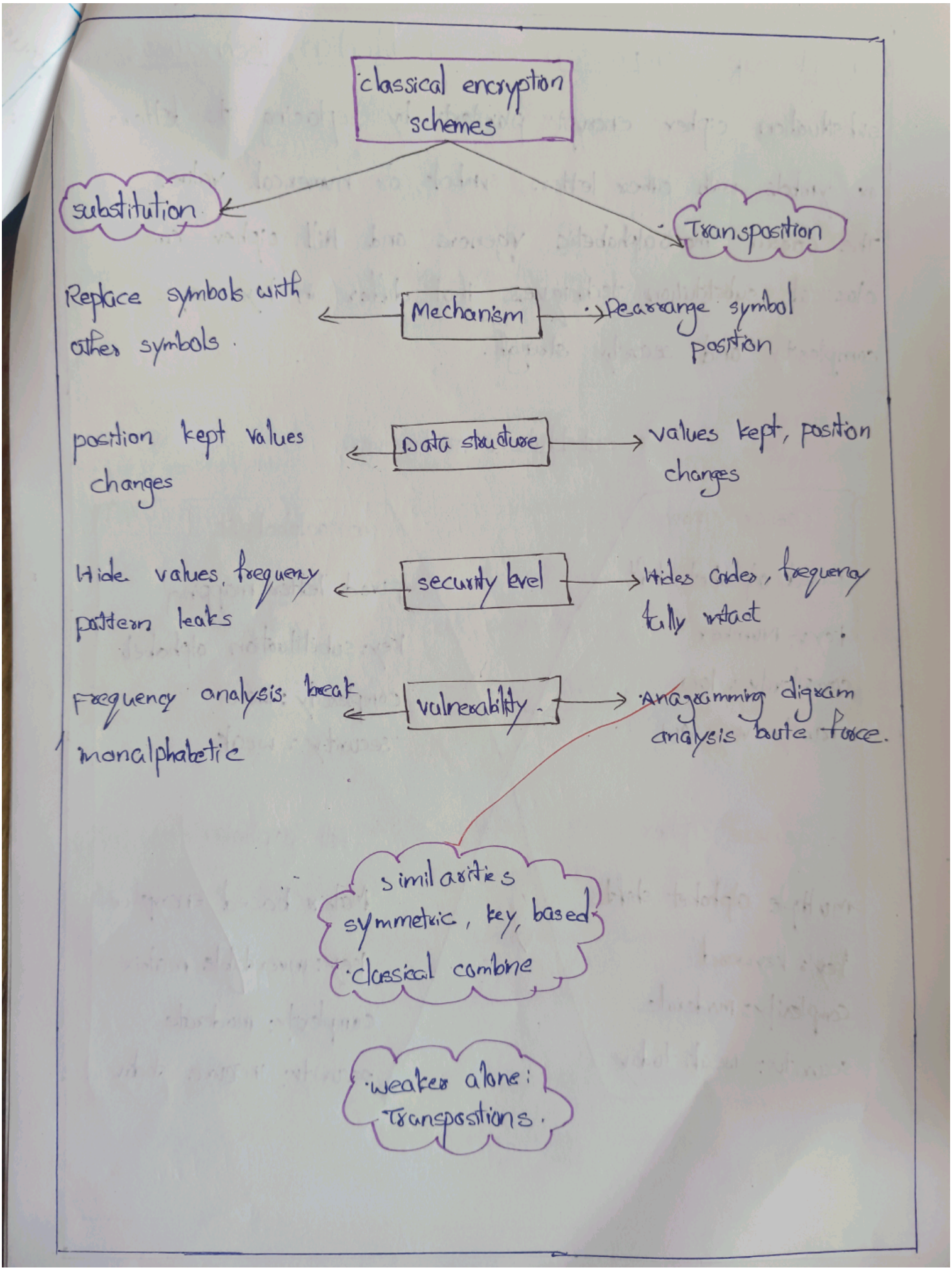
concept map:-





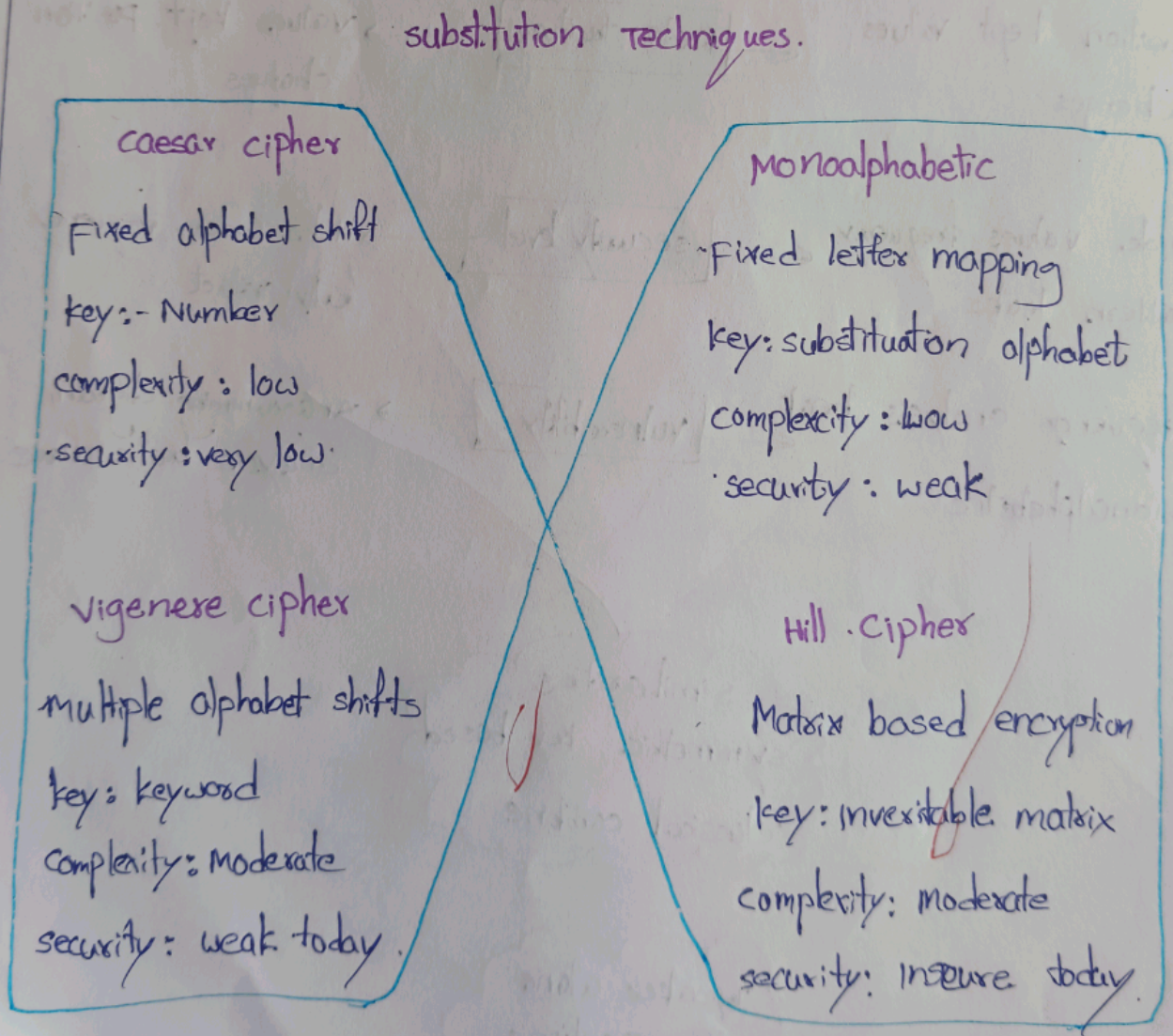
Real world example

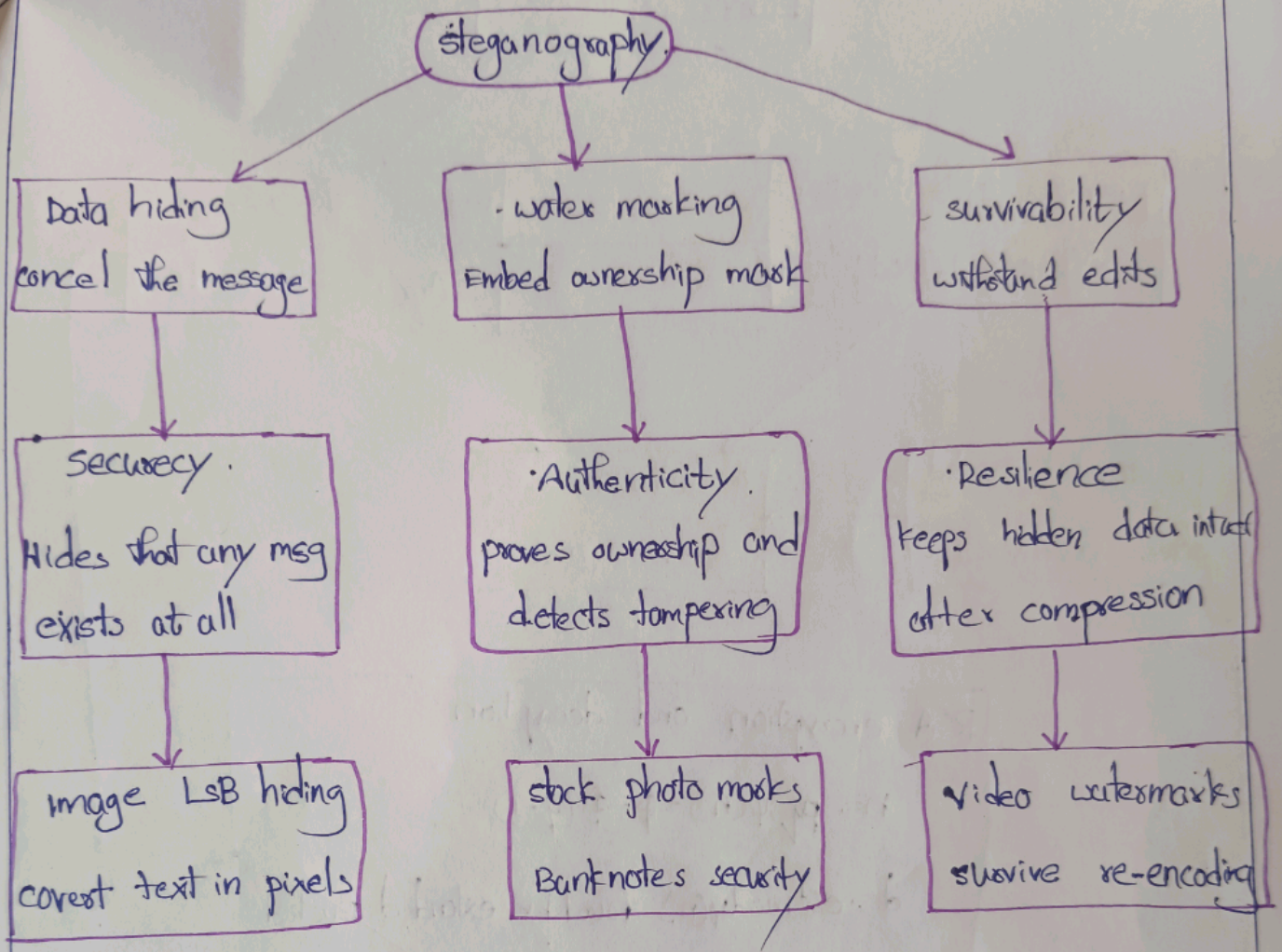
* A fake banking email tricks a customer into revealing login credentials.
* An attacker modifies the amount in an electronic transaction.
* An online shopping website become inaccessible during an attack.
* Ransomware encrypts a company's files and prevents employees from working.


classical encryption schemes
substitution
Transposition
Replace symbols with other symbols.
Mechanism
Rearrange symbol position
position kept values changes
Data structure
values kept, position changes
Hide values, frequency pattern leaks
security level
Hides codes, frequency fully intact
Frequency analysis break monalphabetic
vulnerability
Anagramming digram analysis brute force.
similarities
symmetric, key, based
classical combine
weaker alone:
Transpositions.

concept map - compasison of subsitituation Techniques:-

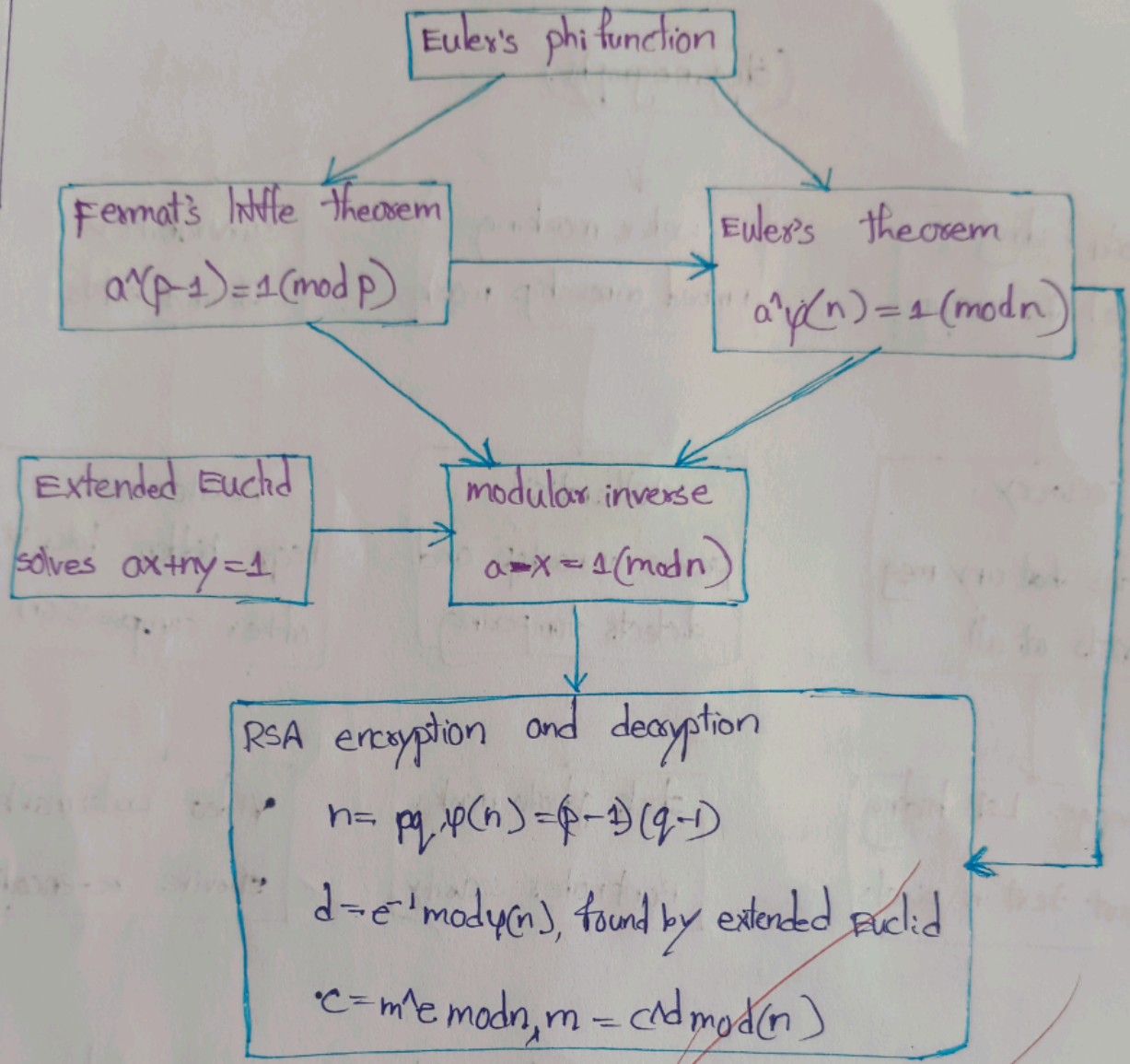
substitution cipher encrypt plaintext by replacing its letters or symbols with other letters, symbols, or numerical values, The Caesar, monoalphabetic, vgenere, and Hill cipher are classical substitution techniques that differ in key usage, complexity and security strength.





An attacker can exploit repeated letters, common words, letters frequencies and a small key space. more complex classified have additional weakness such as repeated key patterns or linear relationships.

The original letters are preserved the ciphertext has exactly the same letter frequency distribution.


Euler's phi function
Fermat's little theorem
$a^{(p-1)} = 1 \pmod p$
Euler's theorem
$a^{\varphi(n)} = 1 \pmod n$
Extended Euclid
solves $ax+ny=1$
modular inverse
$a \cdot x = 1 \pmod n$
RSA encryption and decryption
$n = pq, \varphi(n) = (p-1)(q-1)$
$d = e^{-1} \bmod \varphi(n)$, found by extended Euclid
$c = m^e \bmod n, m = c^d \bmod (n)$